

Chapter 9

Relational Database Design by ER- and EER-to-Relational Mapping

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Chapter Outline

- **ER-to-Relational Mapping Algorithm**
 - Step 1: Mapping of Regular Entity Types.
 - Step 2: Mapping of Weak Entity Types.
 - Step 3: Mapping of Binary 1:1 Relation Types.
 - Step 4: Mapping of Binary 1:N Relationship Types.
 - Step 5: Mapping of Binary M:N Relationship Types.
 - Step 6: Mapping of Multivalued attributes.
 - Step 7: Mapping of n-ary Relationship Types.
- **Mapping Enhanced ER (EER) Model Constructs to Relations**
 - Step 8: Options for Mapping Specialization or Generalization.
 - Step 9: Mapping of Categories (Union Types).

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The ER Conceptual Schema for the COMPANY Database

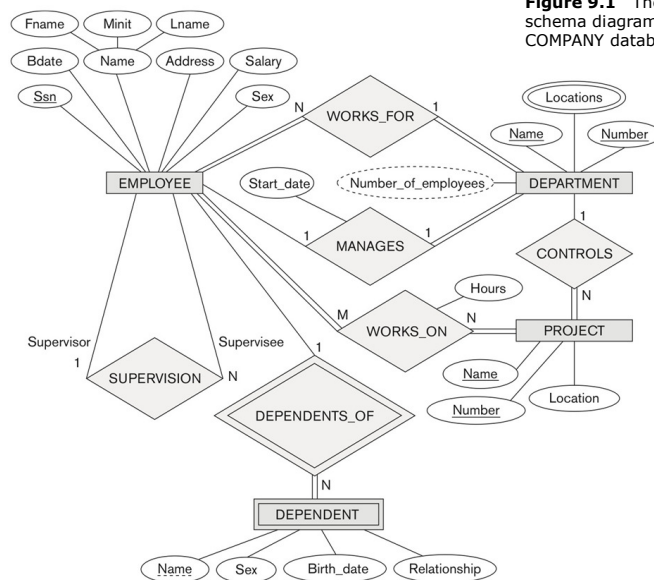


Figure 9.1 The ER conceptual schema diagram for the COMPANY database.

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Result of Mapping the COMPANY ER Schema into a Relational DB Schema.

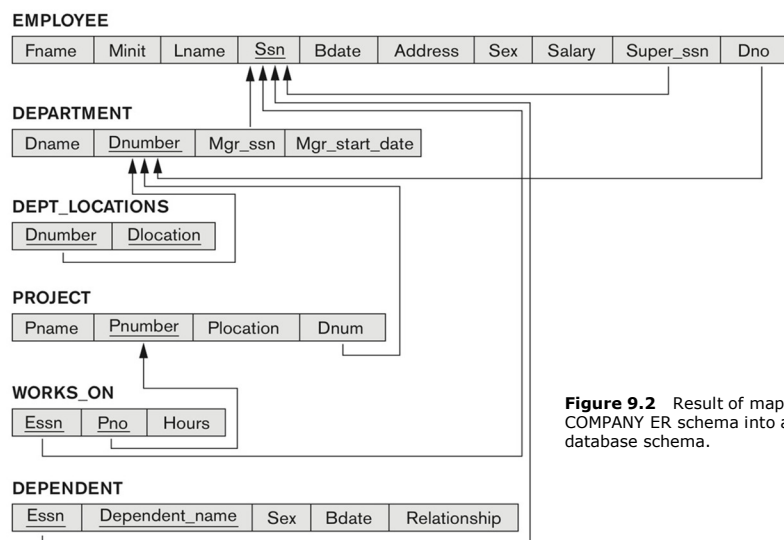


Figure 9.2 Result of mapping the COMPANY ER schema into a relational database schema.

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ER-to-Relational Mapping

- Step 1: Mapping of Regular Entity Types.
 - For each regular entity type E in the ER schema, create a relation R that includes all the simple attributes of E.
 - Store only simple component attributes of each composite attribute.
 - Choose one of the keys of the entity type E as the primary key for R. Remaining keys can be specified as unique keys (aka secondary keys).
- **Example:** We create the relation schemas EMPLOYEE, DEPARTMENT, and PROJECT in the relational DB schema corresponding to the regular entity types in the ER diagram.
 - Ssn, Dnumber, and Pnumber are the primary keys for the relation schemas EMPLOYEE, DEPARTMENT, and PROJECT, respectively.

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Result of Some Mapping Steps

Figure 9.3 Illustration of some mapping steps.

- (a) *Entity* relations after step 1.
 (b) Additional *weak entity* relation after step 2.
 (c) *Relationship* relations after step 5.
 (d) Relation representing multivalued attribute after step 6.

(a) EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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DEPARTMENT

Dname	<u>Dnumber</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation
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(b) DEPENDENT

<u>Essn</u>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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(c) WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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(d) DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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ER-to-Relational Mapping (cont'd)

- **Step 2: Mapping of Weak Entity Types**
 - For each weak entity type W in the ER schema with owner entity type E, create a relation R and include all simple attributes (and only simple components of composite attributes) of W as the attributes of R.
 - The primary key of the relation schema that corresponds to the owner entity type E is included in R as a foreign key.
 - The primary key of R is the combination of the primary key of the owner entity type and the partial key of the weak entity type W.
- **Example:** Create the DEPENDENT relation schema for the weak entity type DEPENDENT.
 - Include the primary key Ssn of the EMPLOYEE relation schema as a foreign key of DEPENDENT (renamed to Essn).
 - The primary key of the DEPENDENT relation schema is the combination of {Essn, Dependent_name} because Dependent_name is the partial key of DEPENDENT entity type.

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ER-to-Relational Mapping (cont'd)

- **Step 3: Mapping of Binary 1:1 Relationship Types**
 - For each binary 1:1 relationship type R in the ER schema, identify the relation schemas S and T that correspond to the participating entity types.
 - Choose one of the two relation schemas, say S, and include the primary key of T in S as a foreign key. It is better to choose an entity type with total (i.e., mandatory) participation in R in the role of S.
 - Include all the simple attributes (and simple component attributes) of the 1:1 relationship type R in S.
- **Example:** 1:1 relationship MANAGES is mapped by choosing the DEPARTMENT relation schema to serve in the role of S, because the participation of DEPARTMENT entities in the MANAGES relationship is total.
 - The primary key Ssn of EMPLOYEE is included in DEPARTMENT as a foreign key (renamed to Mgr_ssn).
 - The attribute of MANAGES, Mgr_start_date, is also included in DEPARTMENT.

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ER-to-Relational Mapping (cont'd)

- **Step 4: Mapping of Binary 1:N Relationship Types.**
 - For each regular binary 1:N relationship type R, **identify the relation schema S that represents the N-side (participating) entity type.**
 - **Include the primary key of the relation schema T that represents the 1-side (participating) entity type as a foreign key in S.**
 - **Include all the simple attributes (and simple component attributes) of the 1:N relation type R as attributes of S.**
- **Example:** For WORKS_FOR relationship type, we include the primary key Dnumber of the DEPARTMENT relation schema as a foreign key in the EMPLOYEE relation schema (and renamed it to Dno).

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ER-to-Relational Mapping (cont'd)

- **Step 5: Mapping of Binary M:N Relationship Types.**
 - **For each regular binary M:N relationship type R, create a new relation schema S to represent R.**
 - **Include the primary keys of the relation schemas that represent the participating entity types as foreign keys in S; and their combination will form the primary key of S.**
 - **Also include all the simple attributes (and simple component attributes) of the M:N relationship type in S.**
- **Example:** The M:N relationship type WORKS_ON is mapped by creating a relation schema WORKS_ON in the relational DB schema.
 - The primary keys of the EMPLOYEE and PROJECT relation schemas are included as foreign keys in WORKS_ON and renamed to Essn and Pno, respectively. The primary key of the WORKS_ON relation schema is the combination of {Essn, Pno}.
 - The Hours attribute of the WORKS_ON relationship type becomes an attribute of the WORKS_ON relation schema.

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ER-to-Relational Mapping (cont'd)

- **Step 6: Mapping of Multivalued attributes.**
 - For each multivalued attribute M, create a new relation schema R.
 - This relation schema R will include an attribute corresponding to M, plus the primary key K of the relation schema that represents the entity type (or relationship type) that has M as an attribute.
 - The primary key of R is the combination of K and M. If the multivalued attribute is composite, we include only its simple component attributes.
- **Example:** A relation schema DEPT_LOCATIONS(Dnumber, Dlocation) is created:
 - The attribute Dlocation represents the multivalued attribute Locations of the DEPARTMENT entity type, while Dnumber (as a foreign key) represents the primary key of DEPARTMENT.
 - The primary key of DEPT_LOCATIONS is the combination of {Dnumber, Dlocation}.

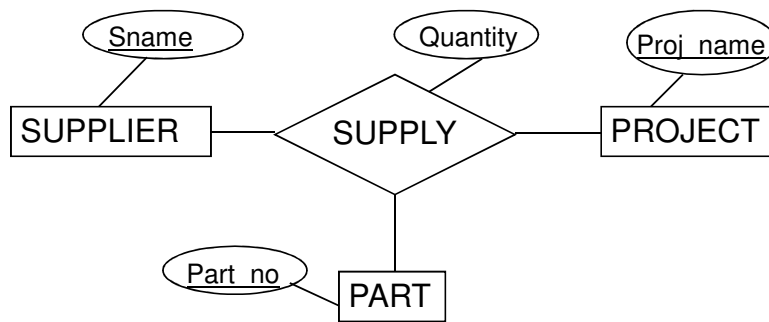
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ER-to-Relational Mapping (cont'd)

- **Step 7: Mapping of n-ary Relationship Types.**
 - For each n-ary relationship type S, where $n > 2$, create a new relation schema R to represent S.
 - Include the primary keys of the relation schemas that represent the entity types participating in S as foreign keys in R; and their combination will form the primary key of R.
 - Also include all the simple attributes (and simple component attributes) of the n-ary relationship type S in R.
- **Example:** The relationship type SUPPLY in the ER model on the next slide.
 - This can be mapped to a relation schema named SUPPLY, whose primary key is the combination of the three foreign keys {Sname, Part_no, Proj_name}.

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Ternary SUPPLY Relationship



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Mapping of the Ternary SUPPLY Relationship

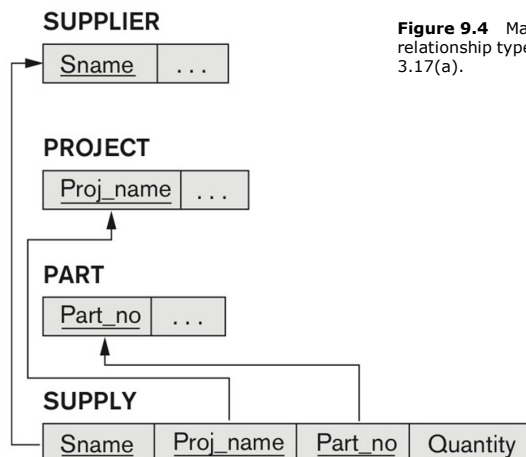


Figure 9.4 Mapping the n -ary relationship type SUPPLY from Figure 3.17(a).

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Summary of Mapping ER constructs and constraints

Table 9.1 Correspondence between ER and Relational Models

ER MODEL	RELATIONAL MODEL
Entity type	<i>Entity</i> relation
1:1 or 1:N relationship type	Foreign key (or <i>relationship</i> relation)
M:N relationship type	<i>Relationship</i> relation and <i>two</i> foreign keys
<i>n</i> -ary relationship type	<i>Relationship</i> relation and <i>n</i> foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key